

CSCI 447 — Machine Learning

Project 2

Assigned: September 21, 2026
Project Due: October 14, 2026

1 Project Description

The purpose of this assignment is to give you some hands-on experience training decision trees for classification and regression. Although we are not going to directly use anything from the module on rule induction; you might want to examine the rules your trees learned to see if they “make sense.” Specifically, you will be implementing a standard univariate (i.e. axis-parallel) decision tree and will compare the performance of the trees when grown to completion to trees that use reduced error pruning.

We need to stress that the number of children of a node in a decision tree is based upon the number of values the feature can take on. For numerical features, you will implement binary splits, even though the content discusses discretization methods. For categorical (nominal and ordinal) features, the number of splits is based on the number of discrete values. When such features exist, then for classification, we choose the feature that maximizes the gain ratio:

$$\text{gainRatio}(f_i) = \frac{\text{gain}(f_i)}{IV(f_i)}$$

where

$$\text{gain}(f_i) = \underbrace{\left(- \sum_{\ell=1}^k \frac{c_{\pi,\ell}}{|\mathcal{D}_{\pi}|} \lg \frac{c_{\pi,\ell}}{|\mathcal{D}_{\pi}|} \right)}_{\mathcal{H}(\mathcal{D}_{\pi})} - \underbrace{\left(\sum_{j=1}^{m_i} \frac{|\mathcal{D}_{\pi}^j|}{|\mathcal{D}_{\pi}|} \mathcal{H}(\mathcal{D}_{\pi}^j) \right)}_{E_{\pi}(f_i)}$$

with $\mathcal{D}_{\pi}$ being the subset of data in partition π , $c_{\pi,\ell}$ denoting the number of examples in partition π that occur in class ℓ , $\mathcal{H}(\mathcal{D}_{\pi})$ is the entropy of partition $\mathcal{D}_{\pi}$, and m_i is the number of domain values in feature f_i . In addition,

$$IV(f_i) = - \sum_{j=1}^{m_i} \frac{|\mathcal{D}_{\pi}^j|}{|\mathcal{D}_{\pi}|} \lg \frac{|\mathcal{D}_{\pi}^j|}{|\mathcal{D}_{\pi}|}.$$

For regression trees, we choose the split that minimizes the mean squared error:

$$\text{Err}'_{\pi} = \frac{1}{|\mathcal{D}_{\pi}|} \sum_j \sum_t (r^t - \hat{r}_{\pi j}^t)^2 b_{\pi j}(\mathbf{x}^t)$$

where r^t is the ground truth response value for $\mathbf{x}^t$, $\hat{r}_{\pi j}^t$ is the predicted response value for $\mathbf{x}^t$ in partition π after following branch j , and

$$b_{\pi j}(\mathbf{x}^t) = \begin{cases} 1 & \text{if } \mathbf{x}^t \in \mathcal{D}_{\pi}^j, \text{ i.e. } \mathbf{x}^t \text{ reaches node } \pi \text{ and takes branch } j \\ 0 & \text{otherwise.} \end{cases}$$

For decision trees, it should not matter whether you have categorical or numeric attributes, but you need to remember to keep track of which is which. In other words, one-hot-coding will not be needed. In addition, as mentioned above, you need to implement the gain ratio criterion for splitting in your classification trees. Make sure to eliminate features that act as unique identifiers of the data points.

2 Data Sets

For this assignment, you will use three classification datasets and three regression data sets that you will download from the UCI Machine Learning Repository or from Canvas, namely:

1. Breast Cancer [Classification]

<https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+%28Original%29>

This breast cancer data set was obtained from the University of Wisconsin

2. Car Evaluation [Classification]

<https://archive.ics.uci.edu/ml/datasets/Car+Evaluation>

The data is on evaluations of car acceptability based on price, comfort, and technical specifications.

3. Congressional Vote [Classification]

<https://archive.ics.uci.edu/ml/datasets/Congressional+Voting+Records>

This data set includes votes for each of the U.S. House of Representatives Congressmen on the 16 key votes identified by the Congressional Quarterly Almanac.

4. Abalone [Regression]

<https://archive.ics.uci.edu/ml/datasets/Abalone>

Predicting the age of abalone from physical measurements.

5. Computer Hardware [Regression]

<https://archive.ics.uci.edu/ml/datasets/Computer+Hardware>

The estimated relative performance values were estimated by the authors using a linear regression method. The gives you a chance to see how well you can replicate the results with these two models.

6. Forest Fires [Regression]

<https://archive.ics.uci.edu/ml/datasets/Forest+Fires>

This is a difficult regression task, where the aim is to predict the burned area of forest fires, in the northeast region of Portugal, by using meteorological and other data .

3 Project Requirements

For this project, the following steps are required:

- Download the six (6) data sets from the UCI Machine Learning repository. You can find this repository at <http://archive.ics.uci.edu/ml/>. The data sets are also available in Canvas. All of the specific URLs are also provided above.
- Implement the decision tree algorithm for classification trees using gain ratio as the splitting criterion.
- Implement the decision tree algorithm for regression trees using mean squared error as the splitting criterion.
- Implement reduced-error pruning (post-pruning) for both the classification trees and the regression trees. Note that you will not be implementing early stopping (pre-pruning) as part of this assignment.
- Run your decision tree implementation on each of the data sets, comparing both pruned and unpruned versions of the trees. These runs should be done with 5×2 cross-validation so you can compare your results statistically.

- You should pull out 20% of the data to be used for pruning and then use the remaining 80% for cross-validation. In other words, during cross-validation, train a decision tree as far as you can on the folds. When pruning, use the 20% that was pulled out to evaluate the node pruning choices. Once the tree is pruned, use the held-out fold for testing.
 - Write a lab report that incorporates the following elements, summarizing the results of your experiments. You can find templates for this format at <http://www.jmlr.org/format/format.html>. The format is also available within Overleaf.
- All project reports must be no more than 15 pages using JMLR format and include the following:
- Title and authors
 - Problem statement with at least one testable hypothesis
 - Experimental approach and explanation of the project design
 - Explanation of the algorithms and how they were implemented (without showing code)
 - Experimental results (tables and/or graphs with explanation)
 - Analysis and discussion of results
 - Two appendices (do not count in page limit):
 - * Main lessons learned
 - * Delineation of who did what on the project
 - Conclusion
 - References (if using any not provided in course content)
- Submit a PDF of your report to the “P2 Paper” portion of the assignment. Canvas is set up to accept PDF files only. Your report will be checked using the TurnItIn plagiarism checking system. The paper must be formatted per JMLR formatting requirements. All math should be included using the math editing capabilities in LaTeX. Figures needs to be high quality. Do not use screenshots.